



Technical Task – Stage 1

Expected completion duration is 7 days after receiving the task.

Your goal is to complete the given tasks within the deadline. You can find the details of the tasks below. At the end, you are expected to prepare a presentation in slides and provide all source codes for review.

Please send the presentation, source codes, and datasets with your name and surname (for zip file) before the deadline for the review. The source codes can be shared via GitHub or as a zip file.

Best of luck.

Task Description: PDF Information Extraction, NLP Analysis & NDS Event Matching

Project Overview

Build a pipeline that:

1. Extracts structured data from drilling daily report PDFs
2. Stores extracted data in a database
3. Matches NDS events from nds_events.xlsx to their source PDFs using text similarity
4. Applies NLP analysis on the extracted text

Task Breakdown

Task 1 - PDF Parsing & Structured Data Extraction

Parse or use OCR all PDFs in PDF_VERSION_1000/ folder and extract the following fields from each report:

Header / Metadata fields:

- Wellbore ID, Report number, Period (start/end datetime), Status
- Operator, Rig Name, Drilling Contractor, Spud Date
- Water depth, Elevation RKB-MSL, Tight well flag, HPHT flag
- Depths (mMD/mTVD): current, kick-off, last casing, formation strength, plug back
- Formation strength (g/cm³), Hole diameter, Pressure test type

Operations Table (one row per operation entry):

- Start time, End time, End Depth mMD
- Main Activity, Sub Activity, State
- Remark / free-text description

Equipment Failure Table (if present):

- Start time, Depths, Equipment system/class, Downtime (min), Remark

Summary Sections (free text):

- Summary of activities (24h)
- Summary of planned activities (24h)

Drilling Fluid Table (if present):

Fluid type, density, plastic viscosity, yield point, sample time/depth

In addition, other sections of the PDFs should be systematically identified and extracted.

Storage: Save all extracted data into a relational or document database

Task 2 - NDS Event Matching

Input: nds_events.xlsx - contains problematic events extracted from the 15/9-F-10 well reports.

Goal: For each NDS event row, find the best matching operation entry in the database (across all PDFs of corresponding well) using text similarity.

The output must include: (i) the matched PDF document, (ii) a similarity score representing the degree of correspondence, and (iii) the matched textual excerpts from the report. The matching methodologies should be benchmarked against used algorithms, and the adoption of creative or advanced matching strategies is strongly encouraged.

Task 3 - NLP Analysis on Extracted Text

Apply the following NLP techniques to **the remark/free-text fields** (operations remarks, summary sections, equipment failure remarks):

3a. Named Entity Recognition (NER) Extract and classify domain-specific entities:

- Depths
- Equipment names (e.g., "FLX packer", "TDS", "BOP", "XO", "spear BHA")
- Measurements (RPM, pressure in bar, flow rate in lpm, weight in MT/ton)
- Time references

3b. Activity Classification / Tagging Train or apply a classifier (even rule-based) to tag each operation remark with a normalized activity label such as: TRIP_IN, TRIP_OUT, CUT, CEMENT, PRESSURE_TEST, EQUIPMENT_FAILURE, REPAIR, WAIT, CIRCULATE, FISHING

3c. Keyword / TF-IDF Extraction For each report, extract top N keywords using TF-IDF across all reports in the corpus. This surfaces what makes each report's language unique - useful for distinguishing well-specific problems.

Note: The implementation of additional data analysis or natural language processing (NLP) techniques would be highly appreciated.